Università di Genova
**DIBRIS** DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

# NLP & SDAI

# A Multi-Agent System for Monitoring Environmental Changes Using Earth Observation Data

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# Creative project

# Difficulty: hard (15.5 pts)

**Objective**

Combine **Symbolic & Distributed AI (S&DAI)** and **NLP** to provide explainable **interpretation** of Earth Observation (EO) data

**Approach**

- Use satellite-derived indicators (wildfires, floods)
- Apply **symbolic reasoning** instead of purely data-driven models
- Produce **structured, human-readable explanations**

**Key Idea**

Bridge **EO data → symbolic reasoning → natural language reports**

**1. EO Layer**

- Indicators extracted from Sentinel-2 imagery
- Examples: vegetation loss, burned area, water increase

**2. Symbolic Multi-Agent System (Reasoning Layer)**

- Implemented in Jason
- *ForestAgent*, *WaterAgent*, *CoordinatorAgent, ExplanationAgent*
- Performs rule-based hazard reasoning

**3. NLP Layer**

- Transforms MAS output into semantic JSON
- Populates OWL ontology
- Generates **final reports** through deterministic NLG

# Motivation

## EO analysis is often a black box

- Data-driven EO systems lack explainability
- Hazard scenarios require explicit reasoning
- Explore feasibility of a symbolic EO pipeline

# System Overview

## A 3-layer architecture

| EO | Symbolic MAS | NLP |
| --- | --- | --- |
| Sentinel-2 data | EO indicators → beliefs | Semantic Transformation (JSON) |
| Indicators extraction | Agent-based reasoning | Ontology population |
| EO evidence generation | Hazard assessment | Report generation |

# EO Layer - Indicators Extraction

## Data

- **Sentinel-2** satellite imagery
- Before/after event comparison
- Change-based indicators
- Population data (ancillary)

## Indicators

- Flood: **NDWI**, water increase
- Wildfire: **NDVI, dNBR**

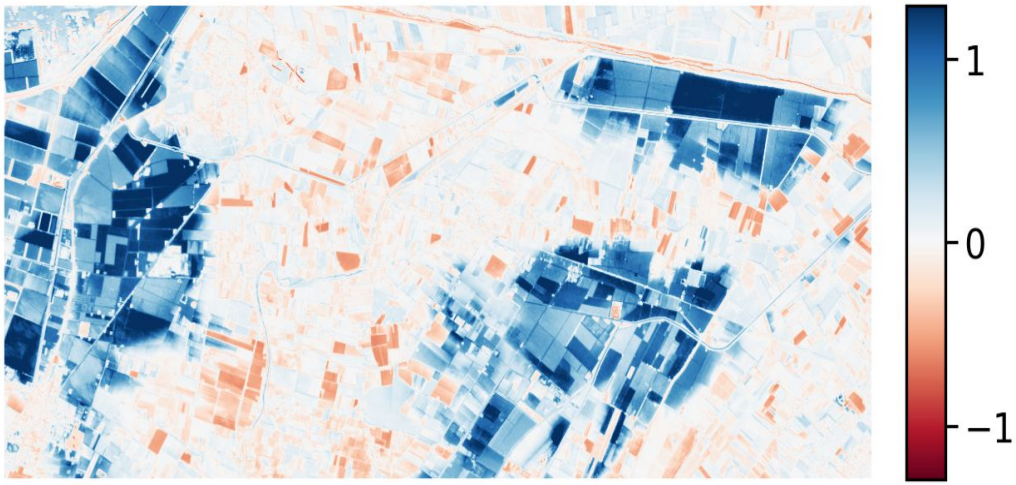
**EO data → structured indicators → input to symbolic reasoning**

# EO Layer - Emilia-Romagna Flood Example

Before

After

NDWI Change




ΔNDWI

# Reasoning Layer

EO indicators → MAS (Agents & Beliefs) → Explanation

## Multi-Agent System (MAS)

**Agents operate over Beliefs Base**

Agents transform EO indicators into semantic assessments using rule-based reasoning

**OUTPUT**
Semantic Explanation

Beliefs and previous layer output (the EO indicators) are added to the beliefs base to drive hazard analysis.

# Reasoning Layer - Beliefs Base

EO indicators → **MAS (Agents & Beliefs)** → Explanation

The beliefs base is composed of:

- EO Indicator beliefs
- Event metadata beliefs
  - Event identity, location, temporal window...
- Context beliefs
  - Population exposure, possible late acquisition...
- Optional user-input beliefs (from the UI)

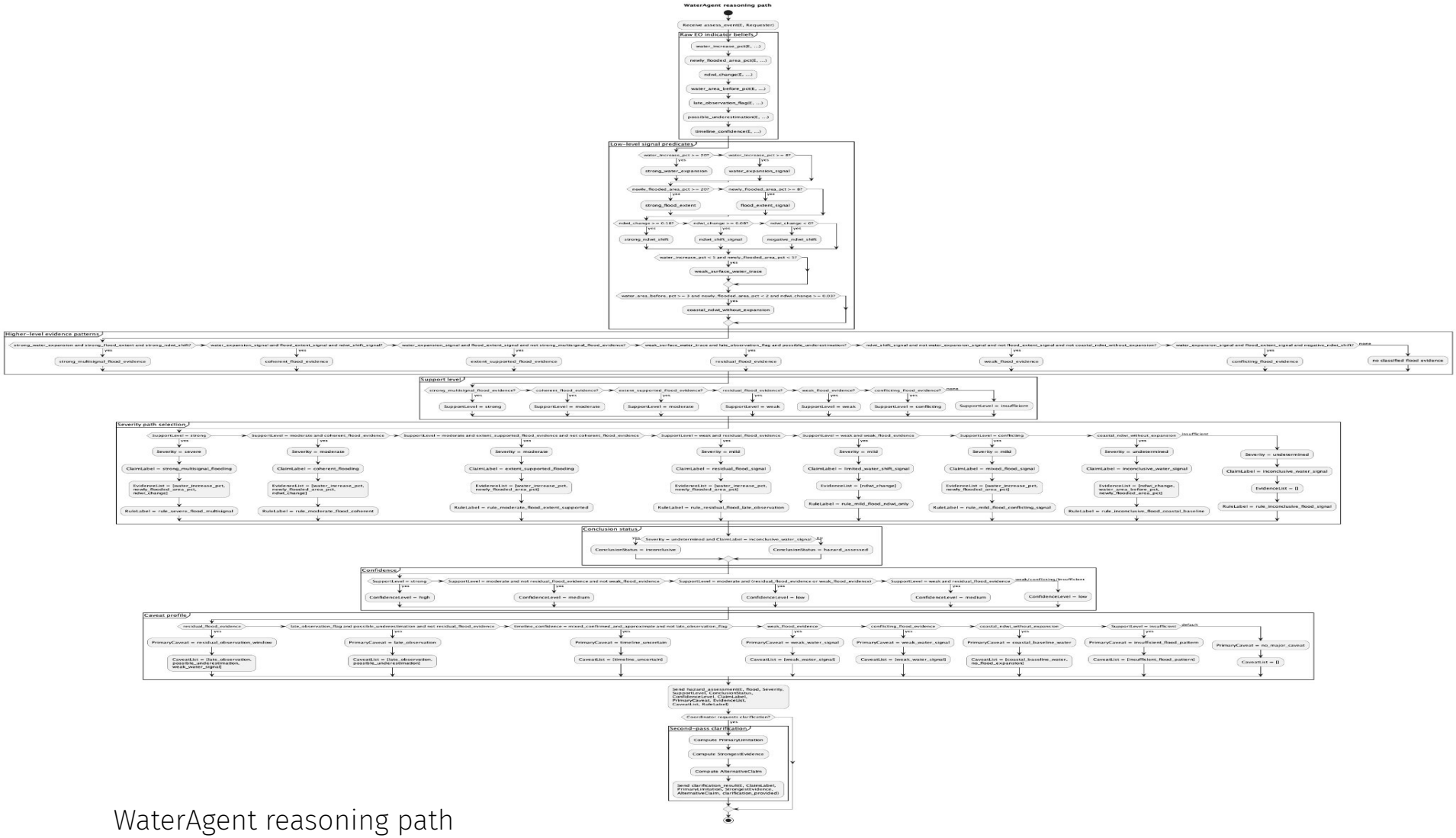
# Reasoning Layer - Agents

EO indicators → **MAS (Agents & Beliefs)** → Explanation

Agents role in the MAS:

- Coordinator agents
  - request hazard assessments and (where needed) clarifications.
  - collect and combines results
- Hazard agents (Water / Forest)
  - produce hazard assessments and transforms EO indicators into semantic concepts using rules
- Explanation agents
  - organise and prepare the results for the NLP layer

# Reasoning Layer- WaterAgent




WaterAgent reasoning path

# Reasoning Layer - Hazard Agents

EO indicators → **MAS (Agents & Beliefs)** → Explanation

1. Read EO and context beliefs
2. Derive indicator-level signals
3. Build higher-level evidence patterns
   - by combining consistent signals
4. Determine support level (how strong the evidence is)
   - weak, medium, strong
5. Map support to severity (how severe the event is)
   - mild, medium, severe
6. Determine conclusion status, confidence and caveats

# Reasoning Layer - Semantic Explanation

**EO indicators → MAS (Agents & Beliefs) → Explanation**

```
semantic_explanation(
  E,
  Event_frame,                  → event metadata
  Assessment_frame,             → severity, confidence, support
  Evidence_frame,               → supporting evidence + caveats
  Clarification_frame,          → limitations, alternative interpretations
  Provenance_frame,             → agent + rule used
  …
)
```

**The semantic explanation encodes both the decision and the reasoning behind it in a structured, machine-readable form.**

Simplified view of the semantic explanation structure

# NLP Layer - From Structure to Language

```
semantic_explanation(
  E,
  Event_frame,
  Assessment_frame,
  Evidence_frame,
  Clarification_frame,
  Provenance_frame,
  …
)
```

Summary
Emilia-Romagna is assessed as a severe flood event with high confidence.

Supporting Evidence
The flood assessment is supported by a strong multi-signal pattern combining water expansion, newly flooded area, and NDWI change.

Caveats
The main limitation is late observation, which may lead to underestimation of the event extent.

Provenance
This assessment was produced by the water agent using the rule `rule_severe_flood_multisignal`.

# NLP Layer - From Structure to Language

## Semantic transformation and explanation delivery

1. Semantic explanation normalisation
2. Ontology population
   - validated, queryable semantic layer (not used for text generation)
3. Natural language report generation

# NLP Layer - Report Generation

Normalisation → Ontology Population → **Report Generation**

Report generation follows a structured NLG pipeline:

- **Content Selection**
  - extraction of relevant semantic parts
- **Lexicalisation**
  - mapping of symbolic labels to natural language expressions
- **Document Planning**
- **Surface Realisation**
  - assembling of the final text

# NLP Layer - Example of a Generated Report

## Generated Report

Raw report text produced from the transformed semantic payload.

```
EO2Explain Event Report: Emilia flood
Location: Emilia-Romagna, Italy
Hazard Type: Flood

Summary
Emilia flood is assessed as a severe flood event with high confidence.

What Happened
The event is located in Emilia-Romagna, Italy. The system classifies it as a severe flood event.

Assessment
The integrated concern level is high, and the case profile is classified as a high priority event. The system treats this assessment
as highly reliable within the available symbolic evidence. Confidence is high because multiple independent EO indicators agree and no
major caveat substantially weakens the interpretation. The interpretation is treated as robust.

Supporting Evidence
The flood assessment is supported by a strong multi-signal pattern combining water expansion, newly flooded area, and spectral water
change. The reasoning path relies on surface-water increase, newly flooded area, and NDWI change.

Caveats And Limitations
No major caveats were attached to this assessment.

Clarification
No second-pass clarification was required because the first-pass specialist assessment was considered sufficient.

Provenance
This assessment was produced by the water agent and grounded in the symbolic rule `rule_severe_flood_multisignal`.

Symbolic Details
- Conclusion status: hazard_assessed
- Severity label: severe
- Support level: strong
- Claim label: strong_multisignal_flooding
- Primary caveat: no_major_caveat
- Clarification status: no_clarification
- Source agent: water_agent
- Rule label: rule_severe_flood_multisignal
```

# Project Extension - Interactive Web UI

- Lightweight interactive web interface
- Upload Sentinel-2 data
- Set contextual assumptions
- Explore explanation reports and reasoning trace

From offline pipeline to interactive and inspectable system

# System Overview

Satellite Data

**EO LAYER**
Earth Observation Processing

**REASONING LAYER**
Multi-Agent Symbolic Logic

**NLP LAYER**
Natural Language Generation

## Generated Report Output

**Summary**

Emilia-Romagna is assessed as a severe flood event with high confidence.

**Supporting Evidence**

The flood assessment is supported by a strong multi-signal pattern combining water expansion, newly flooded area, and NDWI change.

**Caveats**

The main limitation is late observation, which may lead to underestimation of the event extent.

**Provenance**

This assessment was produced by the water agent using the rule `rule_severe_flood_multisignal`

# Evaluation

## Results overview

- **Strong performance on flood events**
  - consistent signals → high confidence assessments
- **More uncertainty in wildfire scenarios**
  - sensitive to EO data quality and timing
- **System behaves conservatively under uncertainty**
  - Lowers confidence and triggers caveats

**Overall: reliable reasoning, but dependent on input data quality**

# Conclusion

- This project integrates EO data with symbolic multi-agent reasoning
- Produces structured and explainable hazard assessments
- Modular and extensible MAS architecture

Strength: transparency and interpretability

Limitation: rule-based design and dependence on EO data quality

# References


Bondarenko et al. (2025) – WorldPop population data

- Bordini, Hübner & Wooldridge (2007)
– Programming Multi-Agent Systems in Jason

- Key & Benson (2006) – dNBR burn severity

- McFeeters (1996) – NDWI index

- Rouse et al. (1974) – NDVI index

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